

Stat4307 Final Exam Question 6

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2023-05-01

Bringing in the information:

```
x<-c(90,86,91,79,83,76,82,83,77,59,88,73,79,84,80,77,89,74,67,83,90,74,68,81,69,73,84,89,72,77)
y<-c(33,50,43,33,21,36,45,18,35,19,49,22,41,60,55,20,55,37,29,21,70,53,13,56,44,32,51,60,19,40)
```

Finding estimates using $n=1800$ (1800 pages in the dictionary)

```
n<-1800
x.est<-mean(x)
y.est<-mean(y)
prob.yest<-y.est/x.est
se.x<-sqrt(var(x)/length(x))
se.y<-sqrt(var(y)/length(y))
se.prob.y<-sqrt((y.est/x.est^2)^2*var(x)+(-1/x.est)^2*var(y))
x.est*n
```

```
## [1] 142620
```

```
y.est*n
```

```
## [1] 69600
```

```
prob.yest
```

```
## [1] 0.4880101
```

```
se.x*sqrt(n)
```

```
## [1] 60.7431
```

```
se.y*sqrt(n)
```

```
## [1] 118.1641
```

```
se.prob.y
```

```
## [1] 0.1984975
```